

Problem 18.3

An annuity-due pays 100 at the beginning of each year forever. The effective annual rate of interest is 25%. What is the present value of the annuity?

Solution

Since the payments are made at the beginning of each year forever, this is a perpetuity-due. The present value of a perpetuity-due with payment R is

$$PV = \frac{R}{d},$$

where d is the effective annual discount rate.

Given

$$i = 0.25,$$

the corresponding discount rate is

$$d = \frac{i}{1+i} = \frac{0.25}{1.25} = 0.20.$$

Therefore,

$$\begin{aligned} PV &= \frac{100}{0.20} \\ &= 500. \end{aligned}$$

Hence, the present value of the annuity is

$$\boxed{\$500}.$$